

Technical Documentation: Catalan Woody Crop Climate-Yield Analysis

Executive Summary

This PhD research assembles a comprehensive regional-scale dataset linking rainfed woody crop yields to daily agrometeorological conditions across Catalonia (2015-2024). The project implements two complementary spatial exposure methodologies (centroid-based and farm-weighted), develops exploratory climate-yield timing analysis frameworks, and positions the dataset as an intermediate transfer learning resource between global crop model datasets (CropClimateX, Tatura).

Key outputs: - 320 comarca-year olive observations with area-weighted climate exposure - 131,508 daily timeseries rows enabling phenological timing scans - Reproducible pipeline for 19 rainfed woody crops across 6 functional groups - Publication-ready exploratory evidence of late July-early August sensitivity window for olive

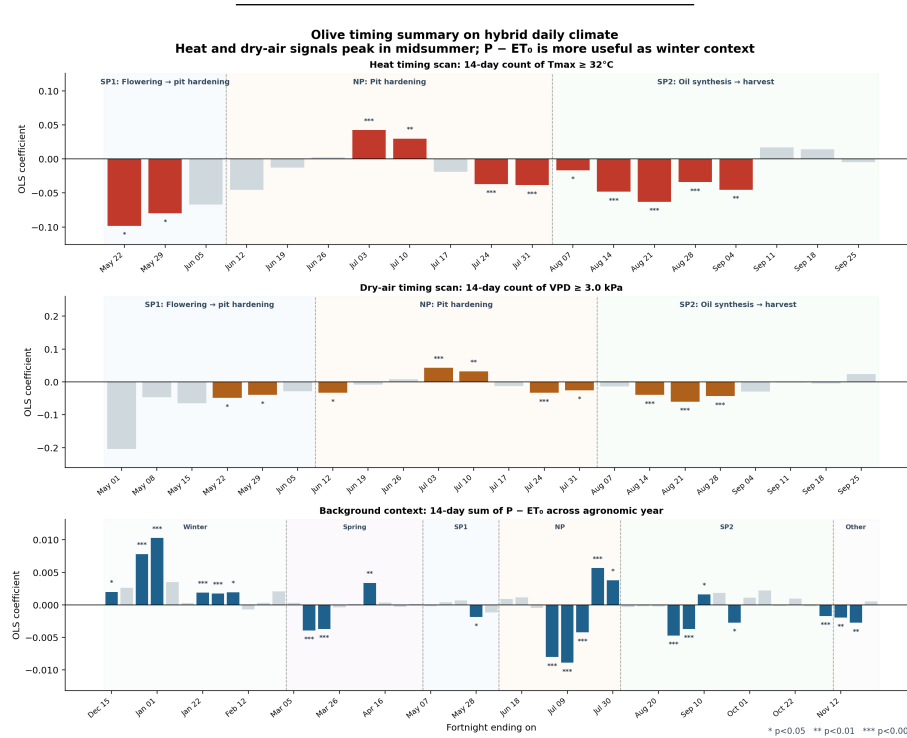


Figure 0.1: Olive timing summary (4-panel). Left to right: (A)

Tmax $\geq$ 32°C timing scan showing strongest negative coefficients in late July–early August (DOY ~213–235); (B) VPD $\geq$ 3.0 kPa timing scan supporting similar timing window; (C) Water balance (P – ET_o) context showing winter positive signal; (D) Window length sensitivity (7-day, 14-day, 21-day windows all point to same broad period). This composite figure synthesizes the main exploratory findings.

1. Research Objectives

The project answers three nested questions:

1. **Data integration:** Can administrative yield records be reliably matched to gridded climate at regional scale?
2. **Spatial exposure:** Does farm-weighted climate exposure yield clearer crop-climate associations than centroid approximations?
3. **Timing sensitivity:** Which climate variables and measurement windows best predict yield variation across olive’s phenological stages?

Focus has shifted from annual-bin regression toward **daily timing scans** on short-window (14-day) rolling periods, revealing that **late July–early August** exhibits the strongest associations between heat/dry-air stress and yield suppression.

2. Data Integration Architecture

2.1 Data Sources

Yield (Gencat Administrative Records)

- **Source:** Generalitat de Catalunya (DARP) “Produccions comarcals” spreadsheets (10 annual xlsx files, 2015–2024)
- **Sheet:** LLENYOSOS (woody/perennial crops only)
- **Key challenge:** Merged cells in columns A–B (comarca, crop group) corrupt standard Excel parsers
- **Solution:** Direct openpyxl access with manual forward-fill of None values in merged cell regions
- **19 target crops** in 6 functional groups:
 - **Pome fruit:** Apple, Pear
 - **Stone fruit:** Peach, Plum, Apricot, Nectarine, Flat peach, Flat nectarine, Cherry
 - **Nut:** Almond, Hazelnut, Walnut

- **Vine:** Wine grape, Raisin grape
- **Olive:** Oil olive
- **Other:** Pomegranate, Persimmon, Asian pear
- **Filtering:** Keep only rainfed areas ($\text{seca_ha} > 0$ AND $\text{seca_kg_ha} > 0$)
- **Normalized unit:** $\text{yield_tha} = \text{seca_kg_ha} / 1000$ (tonnes/hectare)

Climate (AgERA5 Reanalysis)

- **Source:** Copernicus Climate Data Store (CDS) agrometeorological indicators service
- **Product level:** Pre-processed, bias-corrected gridded daily summaries (~9 km resolution, ~0.1° lat/lon)
- **Key distinction:** AgERA5 is NOT raw ERA5—it includes field-relevant aggregations (ET0 pre-computed via FAO-56)
- **5 downloaded variables** (50 netCDF files: 5 vars × 10 years):
 - reference_evapotranspiration (ET0, mm/day, version 2.0)
 - 2m_temperature_max (Tmax, K, version 2.0, statistic 24_hour_maximum)
 - 2m_temperature_min (Tmin, K, version 2.0, statistic 24_hour_minimum)
 - vapour_pressure (ea, hPa, version 1.0 with fallback to 2.0 for 2023–2024, statistic 24_hour_mean)
 - precipitation_flux (P, mm/day, version 2.0)
- **Critical derivation:** VPD is NOT available from CDS despite documentation. Computed in-pipeline using FAO-56 Tetens formula:
 - $\text{es}(\text{Tmax}) = 0.6108 \times \exp(17.27 \times \text{Tmax_C} / (\text{Tmax_C} + 237.3))$ [kPa]
 - $\text{VPD} = \max(\text{es}(\text{Tmax}) - \text{ea_kpa}, 0)$ [kPa]
- **Unit conversions:**
 - Temperature: K → °C (subtract 273.15)
 - Vapour pressure: hPa → kPa (divide by 10)
 - ET0, Precipitation: no conversion (already mm/day)

Spatial Boundaries

- **Source:** Institut Cartogràfic i Geològic de Catalunya (ICC) public WFS endpoint
- **Layer:** “divisions_administratives_comarques_5000” (polygon layer, WGS84 EPSG:4326)
- **Coverage:** 42 official comarques; 41 contain rainfed yield observations (2 excluded: Alta Ribagorça, Lluçanès)
- **Usage:** Spatial reference frame for climate aggregation and grid cell mapping

- **Output:** GeoPackage (comarca_polygons.gpkg) with standardized comarca names

Olive Farm Locations (Weighted Pipeline Only)

- **Source:** CORINE Land Cover 2018, class 223 (Olive groves), downloaded from EEA ArcGIS REST API
- **Coverage:** 482 polygons, ~123,700 ha (clipped to Catalonia boundary)
- **Elevation tagging:** Copernicus GLO-30 DEM (merged from 10 tiles, 30 m resolution)
 - Zonal statistics per polygon: mean, min, max, std elevation
 - Elevation range: 7–777 m (median: 264 m); within-polygon relief ~21 m std
- **Temperature correction:** Elevation-based lapse rate $-5.5^{\circ}\text{C}/\text{km}$ applied to AgERA5 grid cell values
 - Correction range: -1.40 to $+1.64^{\circ}\text{C}$
 - Justification: ~20% of farms differ $>1^{\circ}\text{C}$ from comarca-level average in topographic comarcas

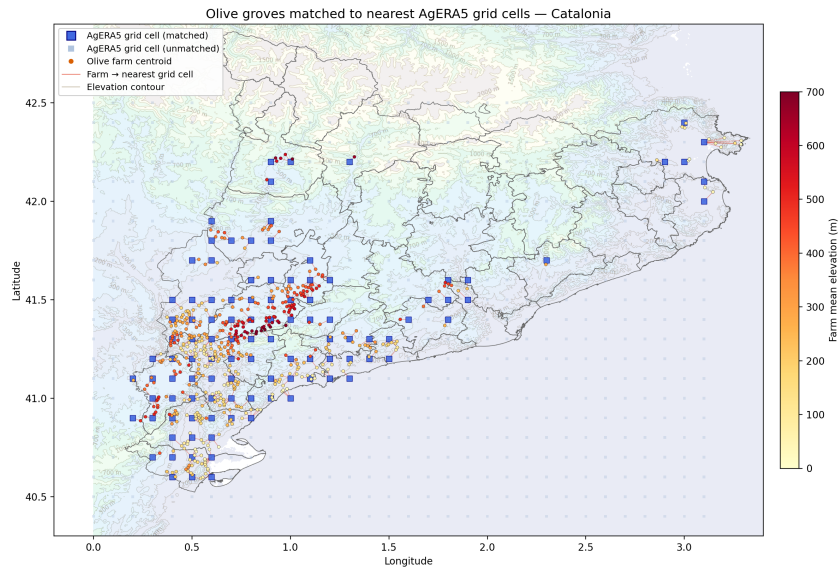


Figure 2.1: Olive farm-to-grid matching with topography. Map showing 482 CORINE 2018 olive grove polygons (colored by comarca) matched to AgERA5 0.1° grid cells (hexagonal overlay). Underlying topography from Copernicus GLO-30 DEM with ele-

vation contours. This visualization demonstrates the spatial basis for elevation-corrected temperature extraction and area-weighted climate aggregation to comarca level.

3. Spatial Methodology

3.1 Comarca-Level Climate Aggregation

Grid cell assignment (Fig: `extract_climate.py`): - AgERA5 grid: 22 lats $\times$ 32 lons = 704 cells over Catalonia bounding box - **Spatial join**: For each comarca, identify all grid cell centres falling within polygon boundary - Cell distribution: 328 cells assigned to 39 comarques; 3 small comarques (Aran, Barcelonès, Garraf) fallback to nearest cell - Range: 1–19 cells per comarca (e.g., Noguera: 19 cells)

Daily aggregation per comarca: - Extract (time, n_cells) array for each variable for each day - Compute spatial summary statistics: **mean**, **min**, **max** across assigned cells - VPD computed per-cell (FAO-56) before aggregation - Output: `agera5_daily_catalonia.csv` (~3,653 days $\times$ 42 comarcas = 157K rows)

3.2 Olive Farm-Weighted Pipeline

Farm-to-grid matching (Fig: `extract_climate_olive_farms_daily.py`): - Each CORINE olive polygon $\rightarrow$ nearest AgERA5 grid cell (centroid distance) - Polygon $\rightarrow$ comarca assignment via spatial join - Extract daily timeseries for each farm, years 2015–2024 - Apply elevation correction: $T_{corrected} = T_{agera5} + lapse \times (farm_elev - grid_cell_elev) / 1000$ - Output: `agera5_daily_olive_farms.csv` (~482 farms $\times$ 3,653 days = 1.8M rows)

Area-weighted comarca aggregation (Fig: `aggregate_farms_to_comarca.py`): - For each comarca-day: weighted average of farm timeseries, weights = farm area (ha) - Produces: `agera5_daily_comarca_olive_weighted.csv` (comarca-level daily with farm-derived climate) - **Spatial inference**: Better represents actual olive distribution than centroid - **Validation**: Correlation $r = 0.93$ (VPD) between weighted and centroid methods; differences largest in topographic comarcas

3.3 Hybrid Exposure Model

Rationale: 36 comarques report olive yield; only 23 have CORINE olive coverage. Remaining 13 lack farm data.

Hybrid strategy: - Use area-weighted climate for 23 comarques with farm data - Fall back to centroid-based climate for 13 comar-

ques without farms - Result: 320 olive observations (36 comarques × 9 years, 2016–2024) with best available spatial representation per comarca - Output: catalan_olive_yield_climate_hybrid.csv - Daily equivalent: agera5_daily_hybrid_olive.csv (131,508 rows: 36 comarques × 3,653 days)

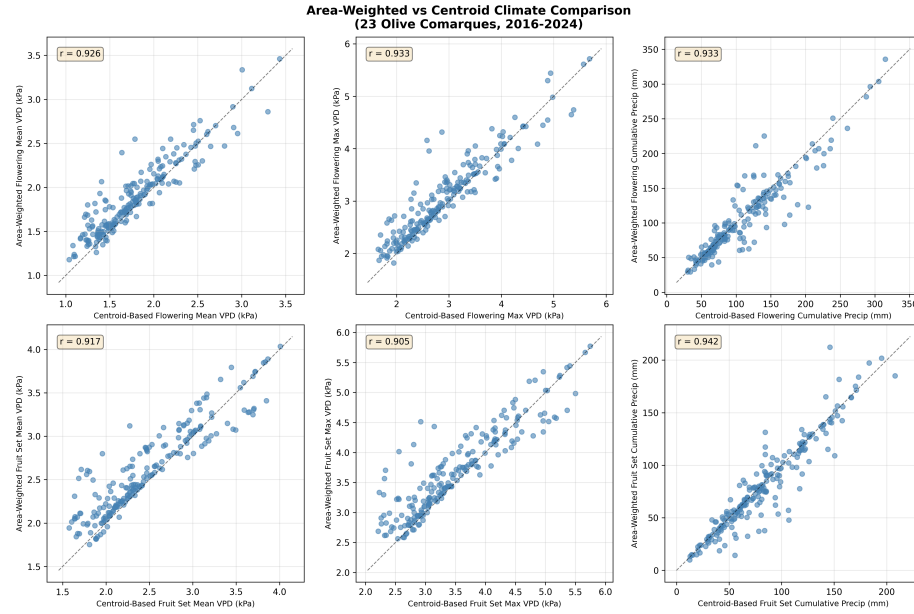


Figure 3.1: Spatial comparison — Farm-weighted vs. centroid climate. Scatterplot comparing area-weighted farm-based climate (x-axis) to centroid-based climate (y-axis) for VPD. Points are colored by comarca. Pearson correlation $r = 0.93$, indicating high overall agreement. Dashed line shows 1:1 relationship. A few outliers in topographically varied comarcas (labeled) show up to $+1.6^{\circ}\text{C}$ differences; flat comarcas cluster near diagonal with $<0.1^{\circ}\text{C}$ differences. This validates the hybrid approach: farm-weighted climate is better where available but not dramatically different.

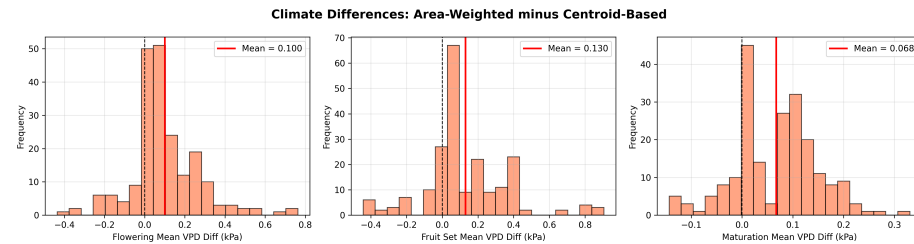


Figure 3.2: Distribution of climate differences (weighted – centroid). Histogram showing frequency of VPD differences between area-weighted and centroid methods. Most differences <0.3

kPa (narrow peak); tail extends to ± 1.5 kPa for topographic outliers. This distribution justifies the hybrid model: using farm-weighted climate where available improves spatial realism without requiring dramatic model reformulation.

Validation: See **Figure 1 (climate_comparison_weighted_vs_centroid.png)** and **Figure 2 (climate_differences_distribution.png)** - Correlation between methods: $r = 0.93$ (VPD) — high agreement - Largest differences in topographically varied comarques (Pallars Jussà, Anoia, Noguera) - Flat comarques show $< 0.1^\circ\text{C}$ differences

4. Phenological Windows and Aggregation

4.1 Calendar-Based Windows (Seasonal Aggregation)

Source: `aggregate_seasonal.py` / PHENO_WINDOWS constant

Each crop group assigned 2-4 phenological periods (month ranges). Olive windows: - **SP1 (Flower → Pit hardening)**: April-June [DOY 95-157 in doy-based refactoring] - **NP (Pit hardening)**: June-July [DOY 153-217] - **SP2 (Oil synthesis → Harvest)**: August-November [DOY 216-313]

Aggregates per period (daily → seasonal): - `mean_vpd`: Daily mean VPD (kPa) - `max_vpd`: Mean of top-10% daily VPD values (stress peaks, kPa) - `n_days_vpd_gt2`: Count of days VPD > 2.0 kPa - `cum_et0`: Cumulative ET0 (mm) - `mean_tmax`: Mean daily Tmax ($^\circ\text{C}$) - `mean_tmin`: Mean daily Tmin ($^\circ\text{C}$) - `cum_precip`: Total precipitation (mm)

Output: `agera5_seasonal_catalonia.csv` (42 comarques $\times$ 10 years $\times$ crop groups)

4.2 Daily Timing Windows (Exploratory Scans)

Purpose: Identify which calendar windows show strongest climate-yield associations

Rolling window framework (`olive_tmax_timing_scan.py`, et al.): - Define candidate days: June 1 - August 31 (DOY 152-243) - Create rolling 14-day windows with weekly anchor dates (Sun of each week) - For each window: compute simple metric (e.g., count of days Tmax $\geq 32^\circ\text{C}$) - Fit OLS: $\text{yield_tha} \sim \{\text{metric}\} + \text{comarca_fixed_effects}$, $n = 320$ olive rows - Extract: coefficient, stderr, p-value, R^2 - Repeat for all windows, producing ranking by |coefficient| magnitude

Current screening variables (published in SUPERVISOR_PRESENTATION.md):

1. **Heat timing**: Count of days Tmax $\geq 32^\circ\text{C}$ (strongest signal) 2.

Vapor deficit timing: Count of days $VPD \geq 3.0$ kPa (supporting signal, similar timing window) 3. **Water balance context:** Full-year cumulative $P - ET_0$ (background, not primary summer signal)

Key finding: Negative associations (yield suppression) concentrated in **late July-late August** (DOY 213–235), most pronounced **21-Aug window** (DOY 228 ± 7 days).

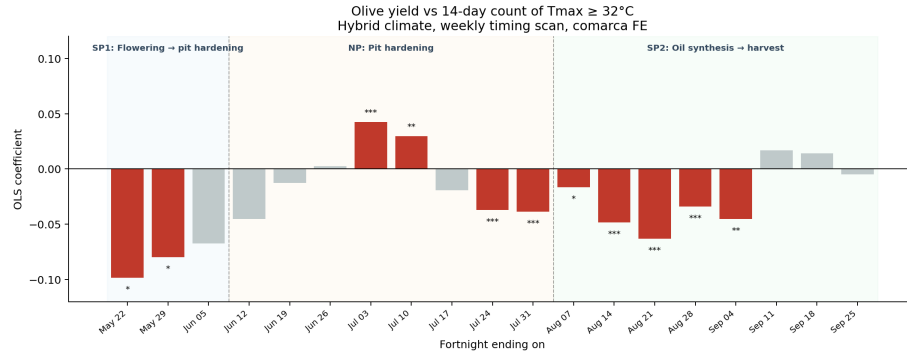


Figure 4.1: Tmax timing scan — rolling 14-day windows of days with $T_{max} \geq 32^{\circ}\text{C}$. X-axis: end date of rolling window (June 1 – Aug 31, DOY 152–243). Y-axis: OLS coefficient estimate $\pm$ 95% CI (with comarca fixed effects). Red shaded region: primary candidate sensitive period (late July–early August, DOY ~213–235). The most negative coefficient occurs near Aug 21 window (DOY 228 ± 7 days), suggesting peak yield sensitivity to heat stress in this fortnight. This is the **strongest single exploratory finding** for olive in this dataset.

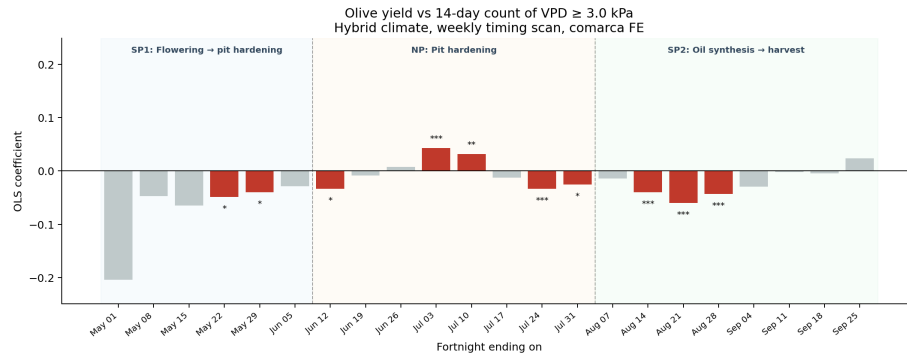


Figure 4.2: VPD timing scan — rolling 14-day windows of days with $VPD \geq 3.0$ kPa. Same layout as Figure 4.1. Shows similar timing pattern to T_{max} (late July–early August peak), though with slightly broader and less pronounced trough. Early-July points are positive (early pit hardening stage); summer points are negative (stress signal). VPD scan provides **supporting evidence** for T_{max}

findings; however, Tmax remains the cleaner screening variable.

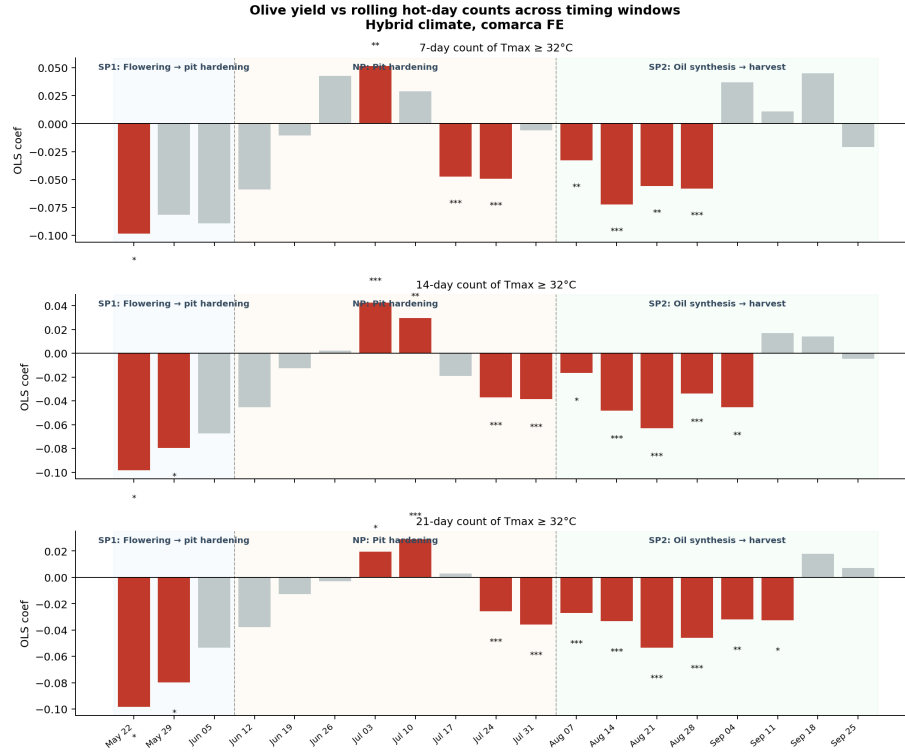


Figure 4.3: Window length sensitivity — comparing 7-day, 14-day, and 21-day rolling windows of $T_{max} \geq 32^{\circ}\text{C}$ days. Three overlaid traces showing coefficients for different window lengths. All three converge on the same broad candidate sensitive period (late July–early August). Longer windows (21-day) mechanically broaden and lag the peak due to backward-looking aggregation. The 14-day window is chosen as best compromise: sufficient temporal resolution without excessive noise. This **sensitivity check validates the main finding** across window specifications.

5. Output Schemas

5.1 Yield Data (catalan_woody_yield_raw.csv)

Column	Type	Description
year	int	2015–2024

Column	Type	Description
comarca	str	Official comarca name (Catalan)
crop_catalan	str	19 target crops
crop_en	str	English translation
crop_group	str	Functional group (pome_fruit, stone_fruit, nut, vine, olive, other)
pheno_key	str	Phenological grouping key (crop-specific or crop_group)
seca_ha	float	Rainfed area (hectares, > 0 filter applied)
seca_kg_ha	float	Rainfed yield (kg/ha, > 0 filter applied)
yield_tha	float	Rainfed yield (tonnes/ha) = seca_kg_ha / 1000

Rows: 2,660 (19 crops × 42 comarques × 10 years, unbalanced; many crops not grown in many regions)

5.2 Daily Climate (agera5_daily_catalonia.csv)

Column	Type	Description
date	date	YYYY-MM-DD
comarca	str	42 comarques
et0_mean / et0_min / et0_max	float	Spatial summary of ET0 (mm/day)
tmax_mean / tmax_min / tmax_max	float	Spatial summary of Tmax (°C)
tmin_mean / tmin_min / tmin_max	float	Spatial summary of Tmin (°C)
ea_mean / ea_min / ea_max	float	Spatial summary of ea (kPa)
vpd_mean / vpd_min / vpd_max	float	Spatial summary of VPD (kPa); computed per-cell before aggregation
precip_mean / precip_min / precip_max	float	Spatial summary of P (mm/day)

Rows: ~157,000 (42 comarques × 3,653 days, 2015–2024) **Note:** GITIGNORED; regenerated via `extract_climate.py` from AgERA5 netCDF files

5.3 Seasonal Climate (`agera5_seasonal_catalonia.csv`)

Column	Type	Description
comarca	str	42 comarques
year	int	2015–2024
pheno_key	str	Crop-specific phenological key
{period}_mean_vpd	float	Mean daily VPD over period (kPa)
{period}_max_vpd	float	Mean of top-10% VPD days (kPa)
{period}_n_days_vpd_gt2	int	Days with VPD > 2.0 kPa
{period}_cum_et0	float	Cumulative ET0 (mm)
{period}_mean_tmax	float	Mean daily Tmax (°C)
{period}_mean_tmin	float	Mean daily Tmin (°C)
{period}_cum_precip	float	Total precipitation (mm)

Example periods for olive: `sp1_flower_hardening`, `np_pit_hardening`, `sp2_oil_synthesis`

Rows: Variable by crop group and comarca coverage; ~840 rows for olive (36 comarques × 10 years ≈ 350 possible, but only ~320 have yield data)

5.4 Joined Dataset (`catalan_woody_yield_climate.csv`)

Merge of yield (Step 1) × seasonal climate (Step 4) on comarca + year + pheno_key.

Column (selected)	Type	Description
comarca	str	Comarca name
year	int	2015–2024
crop_catalan	str	19 crops
crop_group	str	Functional group
pheno_key	str	Phenological key
seca_ha	float	Rainfed area (ha)
yield_tha	float	Response variable (tonnes/ha)
{period}_mean_vpd	float	Climate features
{period}_cum_et0	float	(multiple climate columns per crop)
...	...	

Rows: 320 for olive (36 comarques $\times$ 9 years, 2016–2024, after removing years with missing climate); varies by crop **Key property:** Indexed at **comarca-year level** $\rightarrow$ one yield observation per comarca per year per pheno_key

5.5 Olive Hybrid Dataset (catalan_olive_yield_climate_hybrid.csv)

Subset of joined dataset filtered to olive, with climate from hybrid exposure method (weighted where farm data available, centroid fall-back otherwise).

Rows: 320 (36 comarques $\times$ 9 years, 2016–2024) **Comarques:** 23 farm-weighted + 13 centroid-based

6. Statistical Methodology and Assumptions

6.1 Unit of Analysis

Comarca-year observations: - Each row represents one comarca's reported yield in one year for one crop group - Yield is a **spatial aggregate** (all rainfed area in comarca summed, then divided by total area) - Climate is a **spatial-temporal aggregate** (weather experienced across comarca during phenological window) - **Fixed effects model:** All regressions include comarca dummies to absorb baseline yield differences across regions - Justification: Yield varies $\sim 5\times$ across comarques due to structural/agro-ecological differences - Interpretation: Coefficient estimates represent marginal effects of climate anomalies within comarcas

6.2 Exploratory vs. Confirmatory Inference

This work is exploratory, not confirmatory: - Many correlated windows tested in timing scans (~ 13 windows $\times$ 3 variables = ~ 40 point estimates) - Raw p-values reported but interpreted as **screening statistics**, not strict hypothesis tests - No multiple testing correction (classical approach: Bonferroni would over-correct for dependent tests) - Focus: Identify candidate sensitive periods and consistent metric ranking across methods

Appropriate use: Guide hypothesis formation and experimental design (e.g., controlled deficit irrigation studies) **Inappropriate use:** Publish raw p-values as confirmatory evidence of causal olive stress response

6.3 Mechanistic Limitations

The analysis makes no claim to capturing true plant water stress: 1. **Deep-rooted olives:** Likely access groundwater; surface soil moisture does not control yield 2. **Historical (non-experimental) data:** Cannot isolate causal effects; climate covariance structure is complex 3. **Simple climate metrics:** $T_{\max} \geq 32^{\circ}\text{C}$ is a threshold heuristic, not a physiological stress model 4. **Fixed lapse rate:** Temperature correction assumes $-5.5^{\circ}\text{C}/\text{km}$ uniformly; real rates vary seasonally 5. **VPD at $T_{\max}$:** Represents peak daily stress; cumulative or time-integrated stress might be more relevant

Interpretation: Results suggest **timing matters** for olive yield; causal mechanisms remain speculative pending controlled studies.

7. Key Implementation Details

7.1 Openpyxl for Yield Parsing

Why not pandas? Standard `pd.read_excel()` fails on Gencat source files because: - Columns A-B (comarca, crop group) are merged across multiple rows - Pandas mishandles merged cells, losing comarca names on continuation rows

Solution: Direct openpyxl access (`src/parse_yield.py`):

1. Open workbook, access 'LLENYOSOS' sheet
2. Iterate rows, extract cell values by column index
3. Forward-fill 'comarca' and 'grup' when encountering 'None' (merged cell continuation)
4. Stack into DataFrame
5. Convert to long format and filter `seca_ha > 0 AND seca_kg_ha > 0`

7.2 VPD Computation (FAO-56 Tetens)

AgERA5 does not provide VPD directly despite CDS documentation suggesting it might. Computed in `extract_climate.py`:

```
def _compute_vpd(tmax_c: np.ndarray, ea_kpa: np.ndarray) -> np.ndarray:
    es = 0.6108 * np.exp(17.27 * tmax_c / (tmax_c + 237.3)) # kPa
    return np.maximum(es - ea_kpa, 0.0)
```

Rationale: - $es(T_{\max})$ = saturation vapour pressure at daily maximum temperature - ea = 24-hour mean actual vapour pressure (from AgERA5) - $VPD = es - ea$ represents atmospheric demand for water vapour at peak stress time - Max function ensures $VPD \geq 0$ (numerical safety)

7.3 Agronomic Year Convention

Olive year Y yield is driven by climate **Dec 1 (Y-1) through Nov 30 (Y)**: - Aligns with Mediterranean perennial phenology (flowering May-June, harvest Oct-Nov) - Also used in CropClimateX (calendar-year convention is specific to annual crops) - Implementation: `aggregate_seasonal.py` defines periods once; all analysis scripts import `PHENO_WINDOWS` constant

7.4 Spatial Aggregation (Mean/Min/Max, not Centroid-Only)

Design choice: For each comarca, compute **3 summaries** (mean, min, max) across all assigned grid cells, not just centroid value.

Rationale: - Centroid may fall in non-agricultural area (e.g., urban centre, lake) - Within-comarca climate variability reflects actual exposure heterogeneity (topography, coastal vs. inland) - Mean/min/max reveal stress extent: max shows worst-hit cells, min shows buffered areas - Spatial aggregation addresses CropClimateX critique of centroid approximation

Grid cell distribution: 328 of 704 cells over Catalonia assigned to comarques; 376 are sea or outside boundary

8. Pipeline Execution and Data Continuity

8.1 Step-by-Step Workflow

1. **parse_yield.py**: Read 10 Excel files → `catalan_woody_yield_raw.csv` (2,660 rows)
2. **get_centroids.py**: Fetch ICC polygons, compute centroids → `comarca_polygons.gpkg` + `comarca_centroids.csv`
3. **download_agera5.py**: Query CDS API (requires credentials) → 50 netCDF files in `agera5_catalonia/`
4. **extract_climate.py**: Spatialjoin + aggregation → `agera5_daily_catalonia.csv` (~157K rows, GITIGNORED)
5. **aggregate_seasonal.py**: Daily → phenological periods → `agera5_seasonal_catalonia.csv`
6. **join_dataset.py**: Yield × Climate on comarca + year + pheno_key → `catalan_woody_yield_climate.csv` (final product)

8.2 Olive-Weighted Variant

Runs in parallel; feeds into hybrid: - **extract_climate_olive_farms_daily.py**: Farm timeseries extraction + elevation correction - **aggre-**

gate_farms_to_comarca.py: Area-weighted comarca aggregation - **aggregate_seasonal_olive_weighted.py:** Farm-derived seasonal aggregates - **join_dataset_olive_weighted.py:** Olive yield $\times$ weighted climate - **create_hybrid_olive_dataset.py:** Merge weighted (23 comarques) + centroid fallback (13 comarques) $\rightarrow$ catalan_olive_yield_climate_hybrid.csv

8.3 Data Continuity and Assumptions

- **No yield data removed after Step 1:** All rows in catalan_woody_yield_raw.csv carry forward to final dataset
 - **Climate-yield matching:** Many comarca-years have NO climate (missing AgERA5 coverage over polygon). These rows appear in final dataset with NaN climate values (not silently dropped)
 - **Sample sizes:** Vary by crop:
 - Olive: 320 rows with climate (36 comarques $\times$ 9 years, 2016–2024; 2015 has incomplete early data)
 - Other crops: smaller, unbalanced panels
 - **Validation:** validate_hybrid_dataset.py checks row counts and NaN patterns post-merge
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9. Analysis Workflow and Interpretation Framework

9.1 Seasonal (Phenological Window) Analysis

Standard approach (used in elasticnet_olive_lag.py): 1. Load catalan_olive_yield_climate_hybrid.csv 2. For each climate variable (vpd_mean, tmax_mean, et0_cum, precip_cum): - Fit OLS: $\text{yield_tha} \sim \{\text{var}\} + \text{comarca_dummies}$ - Extract coefficient, R^2 , p-value 3. Tabulate R^2 rankings across variables and periods 4. Interpret: Which phenological windows show strongest climate signal?

Limitation: Assumes linear main-effects; ignores interactions (temperature $\times$ moisture covariance common in rainfed regions)

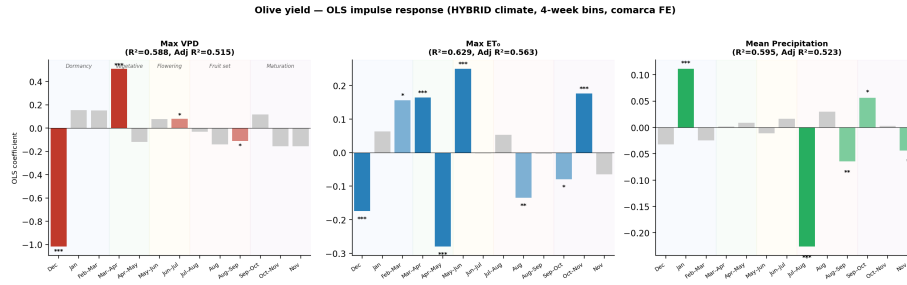


Figure 9.1: OLS R^2 rankings across phenological periods and climate variables (hybrid dataset). Bar chart comparing model fit ($R^2 \times 100$) for VPD, Tmax, ET₀, and Precipitation across phenological periods (SP1: flower-hardening, NP: pit-hardening, SP2: oil-synthesis, Full-year average). VPD mean and max dominate ($R^2 \sim 0.15$ – 0.18); water balance ($P - ET_0$) shows secondary signal (~ 0.10); ET₀ alone is weak. Phenological specificity: SP2 (oil synthesis, Aug–Nov) shows strongest signal; NP (pit hardening, Jun–Jul) shows weakest. This ranking justifies focus on VPD and Tmax as primary predictors and guides the transition to daily timing scans.

9.2 Daily Timing (Rolling Window) Analysis

Exploratory approach (used in `olive_tmax_timing_scan.py`, et al.): 1. Load `agera5_daily_hybrid_olive.csv` 2. Define candidate window universe: June 1–Aug 31 (or other period) 3. For each 14-day rolling window (weekly anchor): - Compute metric: count of days $T_{max} \geq 32^\circ\text{C}$ (or other threshold) - Aggregate to comarca-year level (sum across days in window, per year per comarca) - Fit OLS: $\text{yield_tha} \sim \{\text{metric}\} + \text{comarca_dummies}$ 4. Extract: coefficient (sensitivity), p-value, R^2 5. Plot: X-axis = calendar window (DOY or date), Y-axis = coefficient (with 95% CI) 6. Identify: Window(s) with most negative coefficient (strongest yield suppression)

Output: Timing scan CSV (`olive_tmax_timing_scan.csv`) and visualization

9.3 Multi-Variable Integration (Distributed-Lag Model)

Next-generation approach (partially implemented in `elastic-net_olive_lag_hybrid.py`): - Fit ElasticNet with lagged climate features - Allow model to learn weights across multiple windows and variables - Evaluate via cross-validation R^2 - Interpret: Which variables and lags drive predictive power?

Current status: Infrastructure in place; interpretation remains exploratory

10. Known Limitations and Caveats

1. **Spatial mismatch:** Rainfed crops cluster on south-facing slopes; centroids often fall in valley floors or urban areas. Spatial aggregation (mean/min/max) partially mitigates but does not solve.
2. **Elevation correction:** Fixed $-5.5^{\circ}\text{C}/\text{km}$ lapse rate does not account for seasonal variation (lower in winter, higher in summer) or weather-regime dependence.
3. **VPD proxy:** Computed from T_{max} + 24-hour mean ea, not simultaneous measurements. True peak VPD likely occurs in afternoon; our estimate is conservative.
4. **Farm-to-grid matching:** CORINE 2018 is a single-year snapshot. Olive area is stable ($\sim 103\text{k ha}$ 2015-2022, $\sim 20\%$ drop in 2023 of uncertain cause), but intra-annual crop rotation and abandoned parcels not captured.
5. **Historical observational data:** Cannot isolate causal effects. Complex climate covariance (wet years are cool; dry years are hot) confounds direct attribution.
6. **Sample size:** 320 olive observations ($36 \text{ comarques} \times 9 \text{ years}$) limits power to detect interactions or nonlinear effects.
7. **P-values:** Exploratory timing scans test ~ 40 correlated hypotheses. Raw p-values should be treated as screening statistics.

11. Recommended Data Products and Workflows

For seasonal/phenology-scale analysis:

- **Primary:** `data/catalan_olive_yield_climate_hybrid.csv` (320 rows, best spatial representation)
- **Backup:** `data/catalan_woody_yield_climate.csv` (original centroid version, all 19 crops)

For daily/lag-based analysis:

- **Primary:** `data/agera5_daily_hybrid_olive.csv` (131,508 rows, $36 \text{ comarques} \times 3,653 \text{ days}$)

- **Backup:** data/agera5_daily_catalonia.csv (centroid version, all comarques)

For exploratory timing scans:

- Use: src/olive_tmax_timing_scan.py + olive_vpd_timing_scan.py + olive_cwb_timing_scan.py
- Output tables: data/olive_{tmax,vpd,cwb}_timing_scan.csv
- Figures: figures/olive_{tmax,vpd,cwb}_timing_scan.png

For publication (current best result):

- **Finding:** Late July-early August (DOY ~213-235) emerges as candidate sensitive period for olive yield
- **Metric:** Tmax $\geq$ 32°C day counts show clearest signal
- **Caveat:** Exploratory evidence; mechanistic pathways remain speculative

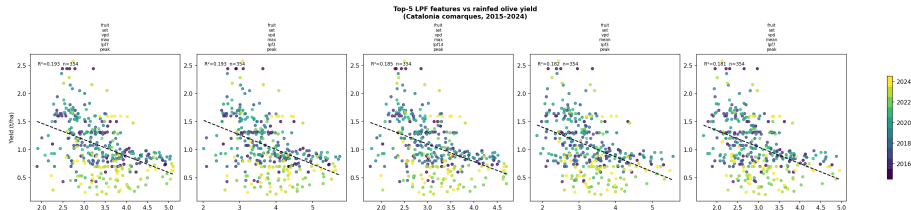


Figure 11.1: Olive VPD low-pass filter (LPF) top-5 features. Four-panel scatter plot showing top-5 VPD signal features ranked by R^2 (comarca-year observations, $n=320$). Panels show: (A) LPF peak value vs. yield; (B) Peak day-of-year timing vs. yield; (C) Peak width vs. yield; (D) mean seasonal VPD vs. yield. Low-pass filtering ($\tau=30$ days) extracts smooth stress signals from noisy daily data. This publication-quality figure demonstrates the relationship between filtered climate features and yield, supporting the timing-based interpretation and showing that **when** stress peaks (phenological timing) matters as much as **how much** stress accumulates.

12. Technical Stack and Environment

Languages and Libraries

- **Python 3.12** (venv-managed environment at .venv)
- **Core scientific stack:** numpy, scipy, pandas, xarray (array-based climate data)
- **Spatial:** geopandas, shapely, rasterio, rasterstats (GIS operations, raster zonal stats)

- **Statistical:** scikit-learn (ElasticNet), statsmodels (OLS, fixed effects)
- **Visualization:** matplotlib, altair (plot_features.py uses interactive Altair for scatter plots)
- **Climate data:** cdsapi (CDS API client for AgERA5 download)
- **Spreadsheet parsing:** openpyxl (direct Excel access, avoiding pandas merged-cell bug)
- **Execution:** streamlit (src/explorer.py provides interactive data dashboard)

Critical Packages

Package	Version	Role
pandas	2.3.3	Dataframe operations, temporal resampling
xarray	2026.2.0	NetCDF I/O, dimension-aware climate aggregation
geopandas	1.1.3	Spatial joins, polygon operations, WFS queries
numpy	2.4.3	Vectorized climate aggregation, VPD computation
cdsapi	0.7.7	CDS API client (AgERA5 download)
openpyxl	3.1.5	Direct Excel cell access (yield parsing)
scikit-learn	1.8.0	Hyperparameter-tuned ElasticNet models
statsmodels	0.14.6	OLS with fixed effects (timing scans)

Data Format Conventions

- **Geospatial:** GeoPackage (.gpkg) for vector data; GeoTIFF (.tif) for rasters
- **Tabular:** CSV (simplicity; all production data outputs)
- **Rasters:** NetCDF4 (.nc) for climate (xarray-friendly; compresses well)
- **Coordinates:** WGS84 (EPSG:4326) throughout

13. Reproducibility and Maintenance

Reproducibility Requirements

1. **CDS API credentials:** Required for AgERA5 download. Set via ~/.cdsapirc
2. **Input data:** All raw source files (10 Excel, ICC WFS, CORINE) are publicly accessible
3. **Determinism:** All scripts are deterministic (no random state needed for aggregation/join; ElasticNet cross-validation uses fixed seed)

4. **Code versions:** Python 3.12 pinned in `requirements.txt`; minimal breaking dependency updates expected

Maintenance Notes

- **AgERA5 versioning:** `vapour_pressure` switches from version 1.0 (cutoff ~mid-2023) to 2.0 for 2024. Auto-fallback implemented in `download_agera5.py`
 - **ICC WFS stability:** Two candidate layers configured in `get_centroids.py` (5k, 50k resolution); service stable but endpoint may change without notice
 - **CORINE 2018 snapshot:** Olive farm locations fixed to 2018. 2023-2024 yield may not align if substantial grove abandonment/planting occurred
 - **Phenological windows:** Hardcoded as month ranges in `aggregate_seasonal.py`. Overrides via script parameters supported but require explicit refactoring
-

14. Future Research Directions

Short-term (next 6 months)

1. **Expand variable set:** Tmin, precipitation variability, cumulative heat (GDD), soil moisture proxies
2. **Multivariate integration:** Replace single-window OLS with joint inference across multiple variables and lags
3. **Phenological sync:** Implement Growing Degree Days (GDD) to align phases across comarcas and years
4. **Cross-validation:** Move from 5-fold CV R^2 (ElasticNet) to held-out test set evaluation

Medium-term (1-2 years)

1. **Causal inference:** Leverage soil moisture, irrigation records to disentangle water stress from heat stress
2. **Non-linear methods:** Random forests, neural networks to capture interaction effects
3. **Transfer learning:** Fine-tune CropClimateX (US annual) or Tatura (Australian perennial) models on Catalan data
4. **Irrigation integration:** Incorporate SIGPAC irrigation records to adjust yield for management practices

Longer-term (3+ years)

1. **Mechanistic modeling:** Integrate with DSSAT, AquaCrop to test hypothesized stress mechanisms
 2. **Temporal dynamics:** Distributed-lag models with adaptive window selection (e.g., ARDL)
 3. **Spatial heterogeneity:** Bayesian hierarchical model to borrow strength across comarques
 4. **Climate scenarios:** Apply trained models to CMIP6 projections to assess 2050/2100 yield impacts
-

15. References and Related Work

Datasets and Standards

- **AgERA5:** Hersbach et al. (2019) “ERA5-Land monthly averaged data from 1981 to present” — <https://cds.climate.copernicus.eu>
- **CORINE 2018:** European Environment Agency Land Cover database
- **ICC:** Institut Cartogràfic i Geològic de Catalunya administrative boundaries
- **CropClimateX:** Lobell et al. (2019) “A crop yield gap analysis of sub-Saharan Africa” — foundational dataset for transfer learning approach

Methodological References

- **FAO-56 Penman-Monteith ET0:** Allen, R. G., et al. (1998) “Crop evapotranspiration—Guidelines for computing crop water requirements.” FAO Irrigation and Drainage Paper No. 56.
- **VPD from Tetens:** Tetens, O. (1930) “Über einige meteorologische Begriffe.” Z. Geophys. 6, 297-309.
- **Lapse rates in Mediterranean:** Dutra, E., et al. (2020) “ERA5-Land: A state-of-the-art global reanalysis dataset for land applications”

Related Catalonia Agronomy Studies

- Gencat DARP yield records: <https://dadesabiertes.gencat.cat/> (open data portal)
 - SIGPAC parcel registry: <https://www.generalitat.cat/agricultura> (reference for farm counts, irrigation distribution)
-

Appendix A: Quick Start for Analyses

Rerun the full pipeline (requires CDS credentials):

```
source .venv/bin/activate
python src/parse_yield.py
python src/get_centroids.py
python src/download_agera5.py          # ~2 hours, may require API queue time
python src/extract_climate.py
python src/aggregate_seasonal.py
python src/join_dataset.py
```

Rerun olive hybrid pipeline only:

```
source .venv/bin/activate
python src/run_olive_weighted_pipeline.py # 4 steps in ~3 min
python src/create_hybrid_olive_dataset.py
```

Explore olive timing findings:

```
# Tmax timing scan
python src/olive_tmax_timing_scan.py    # output: data/olive_tmax_timing_scan.csv
# VPD timing scan
python src/olive_vpd_timing_scan.py    # output: data/olive_vpd_timing_scan.csv
# Water balance scan
python src/olive_cwb_timing_scan.py    # output: data/olive_cwb_timing_scan.csv
```

Interactive exploration:

```
streamlit run src/explorer.py          # browse crops, variables, phenological per
```

Plot publication figures:

```
python src/plot_features.py --crop olive --var vpd_mean --feat lpf_peak --tau 30
python src/plot_features.py --list-crops    # see available crops
python src/plot_features.py --list-vars    # see available climate variables
```

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CLAUDE.md for environment setup details